

## ASCII MASTER FLOW – PANEL 1/12: Design-to-device rail

DESIGN / IP / EDA -> architecture, verification and tape-out

FOUNDRY / FAB -> transistor and interconnect layers on wafer

DICING -> wafer becomes individual dies

ATMP / OSAT -> assemble, test, mark and package

BOARD / MODULE / DEVICE -> downstream electronics integration

## ASCII MASTER FLOW – PANEL 2/12: Semiconductor business-model matrix

FAB -> physical wafer-manufacturing plant

FABLESS -> designs and outsources wafer manufacture

FOUNDRY -> manufactures to customers' designs

IDM -> designs and fabricates its own products

RULE -> plant type and firm model answer different questions

## ASCII MASTER FLOW – PANEL 3/12: Front-end versus back-end

FRONT END -> deposition, lithography, etching, doping, metallisation

OUTPUT -> processed wafer containing dies

BACK END -> bonding, encapsulation, testing, marking, packaging

ATMP -> process description; OSAT -> outsourced service model

TRAP -> back-end capacity is not front-end fabrication

## ASCII MASTER FLOW – PANEL 4/12: Node and lithography ladder

MATURE / MID-RANGE -> broad automotive, industrial and power uses

DUV -> mature and mid-range lithography route

LEADING EDGE -> rising complexity and capital intensity

EUV -> concentrated equipment and export-control chokepoint

RULE -> verify every project-specific node claim officially

## ASCII MASTER FLOW – PANEL 5/12: Advanced integration map

CHIPLETS -> specialised dies combined as one system

2.5D / 3D -> denser package-level integration

THROUGH-SILICON VIAS -> vertical interconnect route

HETEROGENEOUS INTEGRATION -> combine unlike functions or processes

INSIGHT -> performance gains need not begin with smaller lithography

## ASCII MASTER FLOW – PANEL 6/12: Materials taxonomy

COMPOUND SEMICONDUCTORS -> GaN / GaAs / InP / SiC

POWER / RF / OPTOELECTRONICS -> application families

SILICON PHOTONICS -> silicon optical-interconnect platform

POLICY GROUPING -> may place unlike technologies together

RULE -> scheme nomenclature does not change material physics

## ASCII MASTER FLOW – PANEL 7/12: Fab viability chain

RELIABLE POWER + ULTRA-PURE WATER -> stable processing

GASES + CHEMICALS + EQUIPMENT -> controlled process steps

REPEATED RUNS -> yield learning

RELIABILITY EVIDENCE -> customer qualification

OUTCOME -> inauguration alone is not competitive production

## ASCII MASTER FLOW – PANEL 8/12: Institution and scheme map

MeitY -> parent ministry

ISM -> implementing agency

SEMICON INDIA -> scheme package

DLI -> design-focused channel

SPECS / PLI -> separate complementary electronics incentives



## ASCII MASTER FLOW – PANEL 9/12: Semicon 1.0 to 2.0

SEMICON INDIA 1.0 -> Rs 76,000 crore umbrella

FOUR CHANNELS -> fabs, displays, compound/sensor/ATMP, DLI

15 JUL 2026 -> Semicon 2.0 Cabinet approval

SEMICON 2.0 -> Rs 1,27,500 crore approved outlay

SIX PILLARS -> design, tools/materials, fabs, back end, R&D, talent

## ASCII MASTER FLOW – PANEL 10/12: Facility evidence map

DHOLERA -> fab; later named compound-fab plus ATMP approval

MORIGAON -> officially identified OSAT / ATMP location

SANAND -> officially identified back-end cluster and inaugurations

SURAT -> Suchi Semicon OSAT approval

RULE -> location must travel with project and status verb

MUST REMEMBER: The semiconductor chain separates design/IP, wafer fabrication, compound...

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## ASCII MASTER FLOW – PANEL 11/12: Status firewall

APPROVED -> policy permission and support decision

AGREEMENT / GROUNDBREAKING -> financing or construction milestone

INAUGURATED / PRESENTED -> physical or demonstration milestone

QUALIFIED COMMERCIAL SUPPLY -> separate operating evidence

RULE -> never skip a rung or invent production

CLOSE DISTINCTION: Design is not fabrication, fab is not ATMP, packaging is not mere...

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## ASCII MASTER FLOW – PANEL 12/12: PYQ answer spine

DEFINE -> value chain, fab models and ATMP / OSAT

MAP -> ISM, four Semicon 1.0 channels and Semicon 2.0 pillars

ANALYSE -> tools, utilities, talent, yield and qualification

EVIDENCE -> DHRUV64 and facilities with exact status verbs

QUALIFY -> no project total, node, commercial output or answer key

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Date project claims; state value-chain...

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